

SDG ASSIGNMENT

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Course Code: Slot B-IT0614

Course Name: Machine Learning

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1. Explain how Maximum Likelihood can estimate crop disease probability from agricultural data.

Introduction

Crop diseases can seriously reduce agricultural production and cause food shortages. Early identification of crop disease helps farmers take corrective action and protect crops.

Maximum Likelihood Estimation (MLE) is a statistical method used to estimate unknown parameters of a probability model from observed data. In agriculture, MLE can be used to estimate the probability that a crop is diseased based on factors such as temperature, humidity, rainfall, soil moisture, leaf characteristics, and disease observations.

1. Agricultural Data

Suppose agricultural researchers collect data from different farms:

Temperature	Humidity	Rainfall	Soil Moisture	Disease
30°C	80%	High	High	Yes
25°C	60%	Low	Medium	No
29°C	75%	High	High	Yes
24°C	55%	Low	Low	No

The disease condition can be represented as:

- **1 → Diseased**
- **0 → Healthy**

The collected data is used to build a probability model.

2. What is Maximum Likelihood Estimation?

MLE finds the parameter values that make the observed agricultural data **most likely to occur**.

For example, suppose out of 100 plants:

- 30 plants are diseased.
- 70 plants are healthy.

A simple estimate of disease probability is:

Therefore, the estimated probability of disease is **30%**.

However, when several agricultural factors are involved, MLE can estimate the parameters of a more advanced model such as **logistic regression**.

3. Logistic Model for Disease Probability

Let:

- = temperature
- = humidity
- = rainfall
- = soil moisture

The probability of crop disease can be represented as:

Here:

- means disease is present.
- is the intercept.
- are model parameters.
- represents the agricultural features.

MLE finds the values of these parameters that best explain the observed data.

4. Likelihood Concept

Suppose we have observations:

The likelihood function is:

MLE chooses the parameter values that maximize this likelihood:

Usually, the **log-likelihood** is maximized because it is easier to calculate:

5. Step-by-Step Process

Step 1 – Collect agricultural data

Collect information such as:

- Temperature
- Humidity
- Rainfall
- Soil moisture
- Leaf colour
- Leaf spots
- Crop type
- Previous disease occurrence

Step 2 – Label the data

Each crop is labelled:

- Healthy = 0
- Diseased = 1

Step 3 – Select a probability model

A model such as logistic regression can be selected for predicting whether disease is present.

Step 4 – Apply MLE

MLE estimates the model parameters using historical crop observations.

Step 5 – Calculate disease probability

For a new crop, the model uses its environmental and crop characteristics to calculate the probability of disease.

Step 6 – Make a decision

For example:

- Probability = 0.20 → Low risk
- Probability = 0.55 → Moderate risk
- Probability = 0.85 → High risk

A threshold such as 0.5 can be used:

6. Example

Consider a tomato farm where the measured conditions are:

- Temperature = 30°C
- Humidity = 82%
- High rainfall

- High soil moisture

Historical agricultural data shows that disease occurs frequently under these conditions. After applying MLE to estimate the parameters, suppose the model gives:

Therefore:

Disease probability = 87%

Since 87% is greater than the selected threshold of 50%, the crop is classified as **high risk for disease**.

The farmer can then inspect the plants and take preventive measures.

7. Advantages

1. **Uses historical agricultural data** to estimate disease risk.
2. Provides a **probability rather than only Yes/No output**.
3. Can use multiple environmental factors simultaneously.
4. Helps identify high-risk crops at an early stage.
5. Supports data-driven agricultural decisions.
6. Can help reduce unnecessary pesticide application.
7. Helps improve crop yield and reduce losses.

8. Limitations

1. MLE depends strongly on the quality of the collected data.
2. Incorrect or missing sensor readings can affect predictions.
3. The selected probability model must represent the agricultural problem properly.
4. Large datasets may require computational resources.
5. Unusual weather conditions may produce less reliable predictions.

9. Relation to SDG 2 – Zero Hunger

SDG 2 aims to achieve **Zero Hunger** and improve sustainable agriculture.

MLE-based disease prediction can contribute by:

Agricultural data → MLE model → Disease probability → Early detection → Treatment → Reduced crop loss → Higher food production

Therefore, statistical disease prediction can help farmers protect crops, improve productivity, reduce losses, and contribute to food security.

Conclusion

Maximum Likelihood Estimation provides a systematic method for estimating crop disease probabilities from agricultural data. By using factors such as temperature, humidity, rainfall, and soil moisture, an MLE-based model can estimate the likelihood of disease in a crop. Early identification allows farmers to take preventive action, reduce crop losses, and improve agricultural productivity. Hence, MLE can support the objective of **SDG 2 – Zero Hunger**.

2. Analyse how Naïve Bayes can classify recyclable and non-recyclable waste using sensor data.

SDG 11 – Sustainable Cities

Introduction

Increasing waste generation is a major problem in cities. Proper separation of recyclable and non-recyclable waste can improve recycling efficiency and reduce the amount of waste sent to landfills.

Naïve Bayes is a supervised machine-learning classification algorithm based on **Bayes' theorem**. It can classify waste using sensor information such as material type, moisture, weight, colour, and metal detection.

1. Waste Classification

- The system can classify waste into two classes:
- **Recyclable**
- **Non-recyclable**

For example:

Material	Moisture	Metal Sensor	Weight	Class
Plastic bottle	Low	No	Medium	Recyclable
Glass bottle	Low	No	High	Recyclable
Food waste	High	No	Medium	Non-recyclable
Wet tissue	High	No	Low	Non-recyclable
Metal can	Low	Yes	Medium	Recyclable

2. Sensor Data

A smart waste-segregation system can use sensors such as:

- **Moisture sensor** – detects wet/dry waste.
- **Metal sensor** – identifies metallic materials.
- **Weight sensor** – measures the weight of waste.
- **Colour sensor** – detects colour/material characteristics.
- **IR/proximity sensor** – detects the presence of an object.

These sensor readings are supplied to the Naïve Bayes classifier.

3. Bayes' Theorem

Naïve Bayes is based on Bayes' theorem:

$$P(A | B) = \frac{P(B | A)P(A)}{P(B)}$$

4. Why is it called "Naïve"?

Naïve Bayes assumes that the input features are **conditionally independent** given the class.

For example, it assumes that moisture, weight, colour, and metal detection can be considered independently when calculating the probability of a waste class.

The assumption may not always be completely true, but Naïve Bayes often works effectively for classification problems.

5. Naïve Bayes Classification Process

Suppose a waste item produces these sensor readings:

- Moisture = Low
- Metal = No
- Weight = Medium
- Colour = Transparent

6. Step-by-Step Working

Step 1 – Collect training data

Collect sensor measurements from many waste samples.

Step 2 – Label the samples

Each sample is labelled:

- Recyclable
- Non-recyclable

Step 3 – Calculate prior probabilities

For example, if 60 out of 100 training samples are recyclable:

Similarly:

Step 4 – Calculate feature probabilities

The system determines how frequently each sensor condition occurs in each waste class.

For example:

and

are calculated from the training data.

Step 5 – Apply Bayes' theorem

The probabilities of both classes are calculated for the new waste item.

Step 6 – Select the highest probability

The class with the larger posterior probability becomes the predicted class.

7. Example

Suppose a waste item has:

- Low moisture
- No metal detected
- Medium weight
- Transparent colour

After processing the sensor data, assume the classifier produces:

Class	Probability
Recyclable	0.84
Non-recyclable	0.16

Since:

the system classifies the waste as:

RECYCLABLE

The automated waste-segregation system can then direct the item into the recyclable waste container.

8. System Architecture

The complete system can be represented as:

Waste Item

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Sensors

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Sensor Data Collection

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Preprocessing

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Naïve Bayes Classifier

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Calculate Recyclable/Non-recyclable Probability

↓

Classification

↓

Automatic Waste Segregation

9. Advantages

1. **Simple and fast** classification algorithm.
2. Works well with relatively small training datasets.
3. Can process multiple sensor features.
4. Requires comparatively low computational resources.

5. Suitable for embedded and smart-city applications.
6. Produces probability-based classification.
7. Can be updated using new waste data.
8. Helps automate waste segregation.

10. Limitations

1. The independence assumption may not always be realistic.
2. Poor-quality sensor data can reduce accuracy.
3. Some materials have similar physical properties.
4. Mixed or contaminated waste can be difficult to classify.
5. Sensor calibration is important for reliable results.

11. Relation to SDG 11 – Sustainable Cities

SDG 11 focuses on making cities **inclusive, safe, resilient, and sustainable**.

A smart waste-classification system supports sustainable cities by:

Sensors → Waste detection → Naïve Bayes classification → Automatic segregation → Better recycling → Less landfill waste → Cleaner and more sustainable cities

Proper waste segregation increases the amount of recyclable material recovered and reduces environmental pollution.

Conclusion

Naïve Bayes can effectively classify waste as recyclable or non-recyclable using sensor data such as moisture, weight, colour, and metal detection. The algorithm calculates the probability of each waste class and selects the class with the highest probability. When integrated with an automated waste-segregation system, it can improve recycling efficiency, reduce landfill waste, and support smart-city development. Therefore, Naïve Bayes-based waste classification can contribute to **SDG 11 – Sustainable Cities and Communities**.